Reader Report: Week 12

Thompson Sampling, Latent-Space BO, and Causal Inference from Text

Daniel Hardesty Lewis (`dl3645`) · Spring 2026

Summary

BO completion. Thompson sampling draws $\tilde{f} \sim p(f \mid \mathcal{D}_t)$ and maximizes it, providing the first known frequentist sublinear regret bound for nonparametric GP bandits (Chowdhury and Gopalan, 2017), with no free β_t schedule and natural batch parallelism via independent posterior draws. Latent-space BO inserts a learned embedding before the GP — a learned analog of REMBO that makes Thompson sampling tractable in high dimensions.

Causal inference from text. Egami et al. (2022) establish the core framework: text appears as either a *treatment* or *outcome*, and in both roles the analysis depends on a discovered low-dimensional representation g of the documents. Using the same documents for both discovering g and estimating causal effects creates two distinct problems: an *identification* problem (inducing an Analyst Induced SUTVA Violation, AISV) and an *estimation* problem (overfitting). Their remedy is a train/test split (also called a split-sample design): use a training set for discovering g, then use a disjoint test set exclusively for causal effect estimation. Sridhar and Blei (2022) sharpen the point: causal sufficiency of an embedding is an identification claim, not a modeling choice. Veitch et al. (2020) formalize this: joint fine-tuning to predict treatment and outcome yields an asymptotically causally sufficient embedding $\hat{Z} = \lambda(\mathbf{W})$, conditional on the full text $\mathbf{W}$ sufficing for causal adjustment (cf. Feder et al., 2022).

The LLM turn. Imai and Nakamura (2024) bypass learning a representation from observational data: an LLM generates treatment texts via prompting, and its *internal representations* of those texts are extracted for statistical adjustment, disentangling treatment from confounding features; GPI achieves smaller bias and RMSE than observational baselines. Feldman et al. (2026) extend this with sparse autoencoders (SAEs): identify treatment-encoding features via sparse linear probing, steer latent vectors to generate quasi-counterfactuals, then estimate effects via the R-learner.

Critique

The LLM turn substitutes one untestable assumption for another: "no unmeasured confounding conditional on Z" becomes "LLM representation is *causally sufficient for the target domain*" — potentially worse if pretraining under-represents the institutional context. The invariance literature converts this into a falsifiable diagnostic: if sufficiency holds, causal effect estimates should be *invariant* across subpopulations defined by stylistically irrelevant variation; failure, formalizable via IRM (Arjovsky et al., 2019) or Peters et al. (2016), is direct evidence of conflation. This necessary condition is not derived in any of the assigned papers. The BO thread closes the loop: the SAE feature space of Feldman et al. (2026) is a low-dimensional continuous action space, and Thompson sampling over it is a principled adaptive design policy for discovering which latent textual features shift the outcome — a direct instantiation of causal BO (Aglietti et al., 2021).

Questions

1. Egami et al.'s split-sample workflow halves effective sample size. Is there a Bayesian analog — a posterior over text representations propagated into the causal estimand — that avoids this sacrifice? The marginal likelihood over representation hyperparameters (GP empirical Bayes, Week 11) seems like the natural starting point.
2. The invariance test is a *diagnostic* for LLM causal sufficiency. Does it also yield a training *objective* (minimize IRM loss across stylistic environments), and is that strictly stronger than Veitch et al.'s joint fine-tuning or equivalent under their generative model?

Project Update: Text as Confounder Proxy and Treatment in the Zoning Data

Abstract

The Austin zoning dataset contains public testimony — oral and written comment — associated with each of the 7,074 cases. This week I formalize how text enters the causal model in two distinct roles and map each role to the appropriate method from the readings.

Two Roles for Text in the Zoning Data

Role 1: Text as confounder proxy. A protest petition signals community opposition, but its *content* reveals the community's mobilization capacity, political framing, and institutional sophistication — latent confounders for both the petition-filing decision and the council's vote. Standard covariate adjustment cannot condition on raw petition text. The Veitch et al. (2020) framework applies: learn a causally sufficient embedding $\hat{Z} = g(\text{petition})$ by jointly fine-tuning a pretrained language model to predict (a) whether the case was appealed post-decision and (b) final vote outcome, then include $\hat{Z}$ as a covariate in the GP surrogate for the objective function f.

Role 2: Text as treatment. Whether protest rhetoric appeals to environmental, fiscal, or racial-equity concerns may have a direct causal effect on vote outcome, independent of whether a formal protest petition was filed. The Egami et al. (2022) train/test split applies: (a) use the pre-2022 training set ($n = 4{,}312$) to learn a topic model or cluster the petition text into rhetorical frames; (b) use the post-2022 test set ($n = 2{,}762$) to estimate the causal effect of frame on vote, conditioning on $\hat{Z}$ from Role 1. This separation prevents the Analyst Induced SUTVA Violation (AISV) and overfitting.

LLM Augmentation for the Post-2022 Sample

The change-point identified in previous weeks (2022 municipal election) compounds the text problem: the distribution of petition rhetoric and the council's sensitivity to it may have shifted simultaneously. The post-2022 inference sample is already small ($n = 2{,}762$) and cannot support reliable topic discovery. The Imai and Nakamura (2024) GPI approach is attractive here: rather than discovering the rhetorical frame from observational text, use an LLM to *generate* treatment petitions via prompting (with the framing condition specified), then extract the LLM's internal representations of those texts for statistical adjustment of confounding features.

The required assumption — that the LLM's internal representation captures Austin zoning-specific institutional semantics — is testable via the invariance diagnostic proposed in my critique above: estimate the causal effect of rhetorical frame separately for cases decided by different council committees, and test whether estimates are invariant across committees. Failure of invariance would indicate that the LLM representation conflates framing with committee-specific procedural norms, motivating IRM-style fine-tuning before the GPI estimation step.

Next Steps

(1) Embed petition text using a domain-adapted model (BERT fine-tuned on municipal documents) and perform the invariance test across council committees. (2) Implement the Egami et al. split-sample design for the frame-as-treatment analysis. (3) If the invariance test fails, implement IRM fine-tuning of the embedding as a preprocessing step. (4) Integrate $\hat{Z}$ as a covariate in the GP surrogate from previous weeks and evaluate the change in counterfactual comparison results.

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